

BRATA-SLETTOM, Norway

WMO: 013600

Lat: 61.896N

Lon: 7.896E

Elev: 2178

StdP: 13.58

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.4	(c) -2.7	(d) -9.6	(e) 3.6	(f) -7.3	(g) -4.7	(h) 4.6	(i) -2.2	(j) 27.6	(k) 33.2	(l) 26.7	(m) 32.9	(n) 3.2	(o) 340	(p) 0.756

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	71.6	56.9	68.3	55.6	64.9	54.5	59.2	67.4	57.4	65.2	55.7	62.7	11.3	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
55.8	71.9	60.2	54.0	67.2	59.1	52.3	63.1	58.5	26.7	67.5	25.6	65.4	24.5	63.2	65.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 23.4	(b) 22.4	(c) 16.7	(e) -11.6	(f) 75.6	(g) 6.8	(h) 4.5	(i) -16.5	(j) 78.8	(k) -20.5	(l) 81.4	(m) -24.3	(n) 83.9	(o) -29.2	(p) 87.2		
(4) 23.4	(5) 22.4	(6) 16.7	(7) -10.9	(8) 61.3	(9) 6.3	(10) 2.1	(11) -15.4	(12) 62.8	(13) -19.1	(14) 64.0	(15) -22.6	(16) 65.2	(17) -27.2	(18) 66.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.1	19.9	21.0	25.7	35.0	42.7	49.1	55.0	53.9	45.5	35.8	27.7	21.2		
	DBStd	14.85	11.36	10.97	8.82	6.17	6.15	5.07	4.95	5.29	5.56	6.86	9.03	11.79		
	HDD50	5452	934	813	753	450	237	76	10	22	154	442	669	892		
	HDD65	10541	1399	1233	1218	900	690	477	312	344	586	906	1119	1357		
	CDD50	387	0	0	0	0	12	49	165	143	17	1	0	0		
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH74	24	0	0	0	0	0	0	22	2	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	7.4	7.4	7.4	8.6	8.1	8.3	8.2	7.3	7.7	7.6	6.3	6.2	6.4		
	PrecAvg	22.1	2.4	1.9	1.7	0.9	1.1	1.5	2.0	2.0	1.8	2.1	2.2	2.6		
(15) Precipitation	PrecMax	31.7	6.7	4.9	7.9	2.1	3.1	4.0	3.4	3.6	3.2	5.2	4.8	6.1		
	PrecMin	15.3	0.0	0.0	0.1	0.2	0.3	0.2	0.8	0.2	0.5	0.7	0.2	0.4		
	PrecStd	3.6	1.6	1.3	1.5	0.5	0.6	0.9	0.8	0.8	0.7	1.1	1.4	1.4		
	PrecStd	3.6	1.6	1.3	1.5	0.5	0.6	0.9	0.8	0.8	0.7	1.1	1.4	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	42.5	42.5	43.9	53.5	66.8	71.6	77.3	73.4	62.4	54.5	48.5	44.4	40.9		
	0.4% MCWB	38.3	38.1	38.7	43.8	53.6	56.2	61.2	58.2	53.7	48.0	43.3	40.4	37.0		
	2% DB	39.7	40.0	41.0	49.9	61.0	66.2	72.6	71.0	59.0	50.1	45.2	41.7	38.0		
	2% MCWB	36.0	36.2	36.7	42.4	50.4	52.9	58.0	56.8	52.6	45.5	41.3	38.0	35.7		
	5% DB	37.5	37.6	39.0	47.0	57.5	62.2	69.0	67.9	56.3	47.1	41.0	39.1	36.2		
	5% MCWB	34.2	34.3	34.8	40.6	48.2	51.4	56.2	55.8	50.4	43.9	37.3	35.7	33.4		
	10% DB	34.9	35.3	37.1	44.4	54.0	58.9	65.5	64.5	53.9	44.6	38.2	36.2	33.4		
	10% MCWB	31.9	32.3	33.3	39.0	46.3	49.8	55.0	55.1	48.9	41.9	35.2	33.4	30.0		
	0.4% WB	38.7	38.8	39.4	45.2	55.6	57.4	62.5	60.8	55.9	49.4	44.0	40.9	37.0		
	0.4% MCDB	41.9	42.1	43.0	51.2	63.4	67.8	71.6	70.7	59.4	53.4	47.6	44.1	40.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	36.6	36.6	37.2	42.9	51.3	54.7	59.8	58.9	53.5	46.5	41.1	38.4	35.7		
	2% MCDB	39.6	39.6	40.5	48.6	59.3	63.1	69.1	65.8	57.8	49.7	45.3	41.5	38.4		
	5% WB	34.5	34.6	35.3	41.0	49.2	52.8	58.0	57.1	51.5	44.4	37.7	36.2	33.7		
	5% MCDB	37.5	37.5	38.5	46.2	56.6	60.0	66.2	64.9	55.4	46.6	40.3	39.1	36.2		
	10% WB	32.2	32.5	33.5	39.4	46.7	51.1	56.3	55.6	49.6	42.3	35.5	33.7	30.0		
	10% MCDB	35.0	35.4	36.8	44.1	53.6	57.7	63.5	62.9	53.4	44.7	37.6	36.0	33.4		
(35) Mean Daily Temperature Range	MDBR	8.7	9.1	10.7	11.6	13.2	13.3	13.9	14.2	10.7	7.1	7.5	8.3	8.0		
	5% DB	9.6	9.9	10.3	14.7	19.0	20.8	21.4	21.1	14.0	8.9	10.2	9.8	8.0		
	5% WB	8.3	8.5	7.9	8.9	10.7	10.1	9.3	9.2	8.3	6.3	8.4	8.0	8.0		
	MCDBR	9.7	9.6	9.9	13.3	17.1	17.3	17.4	18.1	11.9	8.8	10.1	9.5	8.0		
	MCWBR	8.4	8.5	8.0	8.4	9.9	9.3	8.6	8.7	8.0	7.0	8.5	8.0	8.0		
	MCWBR	8.4	8.5	8.0	8.4	9.9	9.3	8.6	8.7	8.0	7.0	8.5	8.0	8.0		
(40) Clear-Sky Solar Irradiance	taub	0.209	0.259	0.295	0.328	0.333	0.332	0.344	0.334	0.302	0.280	0.216	0.180	0.180		
	taud	2.044	2.184	2.149	2.126	2.209	2.340	2.382	2.408	2.477	2.350	2.071	1.928	1.928		
	Ebn at Noon	158	226	257	271	280	283	276	269	259	219	150	105	7		
	Edh at Noon	11	21	31	39	39	35	33	30	23	17	10	7	4		
(44) All-Sky Solar Radiation	RadAvg	68	244	650	1206	1539	1564	1465	1126	695	317	96	40	4		
	RadStd	13	53	102	145	158	151	146	104	83	45	17	4	4		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (43 neighbors)	N/A	+1.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page